

Building a Synthetic Parallel Catalan–Spanish Corpus of Literary Texts

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Abstract. This paper describes the design and analysis of a synthetic parallel corpus of Catalan literary texts and their automatic Spanish translations. Starting from plain-text documents containing XML tags retrieved from the Corpus Textual Informatitzat de la Llengua Catalana (CTILC), we built a five-script pipeline that (1) loads and normalizes the raw texts, (2) translates them from Catalan into Spanish using Google’s Gemma3-27B-IT, (3) postprocesses and builds a paragraph-level bilingual corpus, (4) analyzes the distribution of the resulting corpus, and finally (5) performs a comparative linguistic analysis covering keyword extraction, n-gram frequencies, and lexical richness metrics. The resulting corpus¹ contains 63 documents (1,577,899 Catalan words; 1,669,666 Spanish words; 30,032 aligned paragraph pairs) published between 1905 and 1948 and spans five dialectal variants of Catalan. The results shed light on the translation behavior of a large language model in literary prose: higher hapax rates in Catalan than in Spanish, broadly comparable MATTR scores across languages, a mild lexical normalization relative to the Catalan source texts regarding diachronic forms, and a slight tendency for translations to be longer, while preserving many named entities.

Keywords: corpus linguistics · parallel corpus · Catalan · Spanish · machine translation

1 Introduction

Parallel corpora, defined as collections of source texts aligned with their corresponding translations (Lefer, 2020), are a fundamental resource for translation studies, contrastive linguistics, and the training and evaluation of machine translation (MT) systems. For less-resourced language varieties such as Valencian and the broader family of Catalan dialects, high-quality parallel resources remain relatively scarce, particularly within the literary domain.

This study addresses that gap by constructing a synthetic parallel Catalan–Spanish corpus from literary texts published between 1905 and 1948. The texts were retrieved from the Corpus Textual Informatitzat de la Llengua Catalana (CTILC) and translated automatically into Spanish using Google’s large language model (LLM) Gemma3-27B-IT (Gemma-Team, 2025).

Although the present study focuses on corpus construction and descriptive linguistic analysis rather than MT training experiments, the parallel alignment and offset traceability of the corpus are designed with downstream MT use in mind. A preliminary expectation of this work is that LLM-generated translations would tend to normalize vocabulary and, therefore, exhibit lower lexical richness than the source texts.

2 Related Work

The scarcity of human-translated data for low-resource languages has led to an increasing reliance on automatic translation as a strategy for corpus construction. Edunov et al. (2018) show that back-translated monolingual data yields substantial improvements in MT quality. Subsequent work has shown that LLM-generated synthetic data can improve MT performance even when the output contains noise (de Gibert et al., 2025), directly supporting the viability of the approach adopted in this study.

¹ All code and data are made publicly available at <https://github.com/guerreropaula/synthetic-corpus-ca-es>.

For Catalan specifically, resources such as the OpenSubtitles corpus (Lison and Tiedemann, 2016) and the broader OPUS collection provide wide-domain parallel data, but literary-domain coverage remains limited. Perez-Ortiz et al. (2024) criticize the persistent gap in parallel resources for Iberian co-official languages, including Catalan, in the context of large-scale MT evaluation.

The use of instruction-tuned LLMs for corpus construction is a relatively recent development. Unlike dedicated transformer-based models such as MarianMT (Junczys-Dowmunt et al., 2018), instruction-tuned LLMs such as Gemma have been shown to produce more fluent and idiomatic translations on literary and domain-specific text, at the cost of reduced control over output format and potential hallucination. The lexical and quantitative analysis presented here contributes empirical evidence on the translation behavior of Gemma3-27B-IT on Catalan literary prose, an underrepresented domain in standard MT benchmarks.

3 Data Sources and Format

3.1 Source Corpus

The raw texts in Catalan were drawn from the CTILC (Corpus Textual Informatitzat de la Llengua Catalana)², a diachronic corpus developed by the Institut d’Estudis Catalans (IEC) covering works between 1832 and 1988. The corpus is distributed as individual files in `.txt` format, although they follow an XML-like structure:

Listing 1.1. Structure of a CTILC source file.

```
<DOCUMENT>
  <OBRA id="3150">
    <AUTOR>Morató i Grau, Josep</AUTOR>
    <TITOL>El tresor</TITOL>
    <ANY>1908</ANY>
    <CLASSIFICACIO_TEXTUAL llengua="LIT"
      traduccio="0" variant="central" />
  </OBRA>
  <TEXT>Full text as a single block...</TEXT>
</DOCUMENT>
```

The metadata embedded in the XML header includes the work’s identifier (`OBRA id`), author name (`AUTOR`), title (`TITOL`), year of publication (`ANY`), textual domain (`llengua`), a binary tag indicating whether the text is a translation (`traduccio`), and a dialectal variant label (`variant`). The full text is stored in the `TEXT` field as a single continuous block. The five dialectal variants present in the collection are *central* (Central), *valencià* (Valencian), *nord-occidental* (North-Western), *baleàric* (Balearic), and *septentrional* (Northern).

3.2 Temporal Scope and Corpus Size

Only documents with a publication year of 1905 or later were retained for translation and subsequent analysis. By filtering out earlier texts, we reduce the impact of orthographic and lexical variation throughout the years, which would otherwise introduce noise and make the machine translation (MT) process more difficult.

The total size of the corpus is 348 files. However, only 98 satisfied the 1905 filtering criterion. Of these, 63 files (totaling 1,577,899 words)³ were ultimately selected for translation due to computational constraints. Appendix A lists the selected works.

² The corpus can be accessed and downloaded at <https://ctilc.iec.cat/>. All works included in this study belong to the CTILC collection; none of the source data is original to the present authors. The selected works have entered the public domain and are available for private and research use under a Creative Commons license.

³ This word count corresponds to raw counts before postprocessing and stopword removal.

4 Pipeline Description

The process followed to obtain the corpus and to perform the subsequent analysis is divided into five Python scripts (contained in `scripts/`), numbered in execution order. Figure 1 summarizes the workflow.

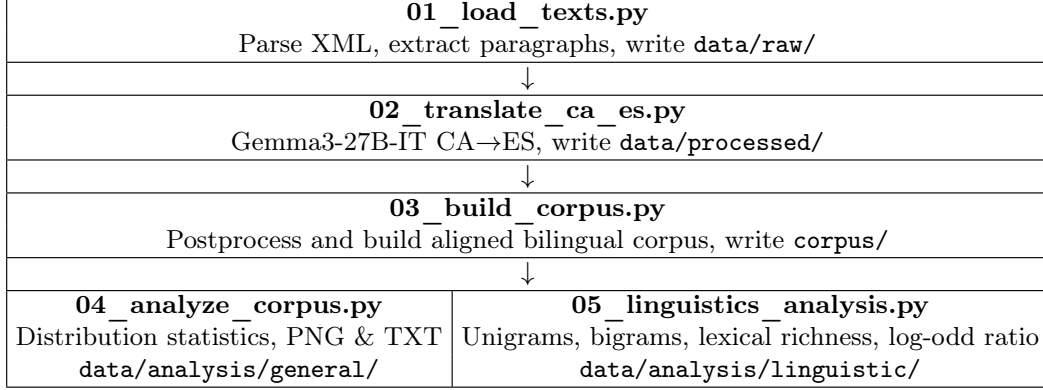


Fig. 1. Overview of the five-script pipeline, covering (1) data extraction from CTILC files and preprocessing, (2) automatic Catalan–Spanish translation using Gemma3-27B-IT, (3) construction of the aligned bilingual corpus with MT artifact removal, and (4 and 5) corpus distribution and linguistic analysis.

4.1 Stage 1: Text Loading (`01_load_texts.py`)

The first script iterates over all `.txt` files in the input directory, `data/corpus_ctilc`, and processes each file as follows.

Metadata extraction. Regular expressions are used to extract the document identifier (`<OBRA id>`), author name (`<AUTOR>`), title (handling both accented and unaccented variants) (`<TITOL>`), and year of publication `<ANY>`, as well as the attributes associated with the textual classification (`<CLASSIFICACIO-TEXTUAL>`), namely the text domain (`llengua`), whether the document is an original or a translated text (`traduccio`), and the dialectal variant (`variant`).

Text extraction and segmentation. The plain text enclosed in `<TEXT>...</TEXT>` is extracted and split on double newlines (`\n\n`) to produce paragraph-level segments. Segments shorter than 30 characters are discarded to eliminate chapter headings and other potential noise. Finally, whitespace within each paragraph is normalized by replacing consecutive whitespace characters with a single space.

Output. Each document is saved as a plain-text file in `data/raw/<doc_id>.txt`, where the `doc_id` has a six-digit identifier derived from the `OBRA id` field (e.g. `doc_000137`). Additionally, a `metadata.csv` file is generated with one row per document. This file stores all extracted metadata fields together with the document IDs and the final paragraph count (`n_paragraphs`).

4.2 Stage 2: Machine Translation (`02_translate_ca_es.py`)

The second script reads each plain-text document from `data/raw/<doc_id>.txt` and translates it into Spanish using Gemma3-27B-IT. Translation is performed at the paragraph level to ensure that inputs remain within the model’s context-length limitations. The resulting translations are written to `data/processed/<doc_id>_es.txt`, maintaining the same paragraph segmentation and document ID as the source.

The result is therefore two aligned textual versions per document: the original Catalan text and its Spanish counterpart, both identifiable via their shared document ID. The total computing time required to construct the corpus was approximately 100 hours.

4.3 Stage 3: Corpus Construction (03_build_corpus.py)

The third script reads the source texts from `data/raw/` and the corresponding Spanish translations from `data/processed/` and builds the parallel aligned corpus. As a preliminary step, the script removes artifacts left by the translation prompt—lines matching a pattern similar to `[Textoliterario. ...]` or an equivalent in English—from all Spanish translation files before alignment begins. A `--dry-run` flag allows inspection of which lines would be removed without modifying any file, and a `--no-preprocess` flag skips this cleaning step entirely.

Alignment is performed at the paragraph level by splitting texts on blank lines. When the Catalan source and Spanish translation have different paragraph counts, semantic alignment is performed using the Multilingual E5 sentence encoder (Wang et al., 2024). Paragraph embeddings are computed for both sides and the algorithm finds the optimal alignment, allowing many-to-one paragraph merging with a gap penalty of 0.2. Character offsets within the original files are recorded for each aligned segment.

Two output files are created in the `corpus/` directory: `corpus_ca_es.csv` (paragraph-level alignment, one row per aligned segment) and `corpus_ca_es_fulltext.jsonl`, a document-level corpus in which paragraphs are joined. The document-level corpus is stored in JSONL rather than CSV because long full-text fields can exceed CSV limits and are often truncated or improperly rendered in standard spreadsheet software.

Additionally, the script filters `data/raw/metadata.csv` to keep only those documents for which a Spanish translation was successfully produced and aligned in `corpus/corpus_ca_es.csv`. The resulting file is stored in `corpus/metadata_filtered.csv` and serves as a registry of documents translated in the present study and included in the final corpus.

4.4 Stage 4: Corpus Distribution Analysis (04_analyze_corpus.py)

The fourth script analyzes the corpus using `data/raw/metadata.csv` and restricts the input to documents for which a Spanish translation exists in `data/processed/`⁴. It performs a set of descriptive statistics summarized as follows.

- **Temporal distribution:** document counts and average paragraph counts by decade.
- **Textual domain and variant distribution:** counts by the `lang` metadata field (domain) and by the `variant` field.
- **Translation status:** proportion of texts tagged as translations (`traduccio = 1`) versus original texts (`traduccio = 0`).
- **Author coverage:** top 15 authors by document count.
- **Word and paragraph counts:** total word count computed over the original Catalan texts and Spanish translations by splitting on whitespace; paragraph counts (total, average, and median per document) are read directly from the `n_paragraphs` column in `metadata.csv`. Additionally, the script reads the aligned `corpus/corpus_ca_es.csv` to report the total number of aligned paragraph pairs and the corresponding token counts for each language.

Output is written to `data/analysis/general/run_*/` and includes a plain-text summary report (`corpus_stats.txt`) and several PNG figures and charts for a more visual analysis. The main modes supported are `--only-processed` (restrict to documents for which a Spanish translation exists), `--year` (filter by minimum year of publication), and `--variant` (filter by dialectal variant).

4.5 Stage 5: Linguistic Analysis (05_linguistics_analysis.py)

The fifth script performs the core corpus-linguistic analysis and supports three modes: single-language analysis of the Catalan originals (`--lang ca`), single-language analysis of the Spanish translations (`--lang es`), and a comparative mode (`--compare`) that runs both analyses and produces additional metrics for the language pair.

⁴ Delimited by two arguments in the conducted experiments: `--year 1905` and `--only-processed`.

Tokenization and Stopwords. Tokenization is adapted for each language. For Catalan, the regex `\b[a-záéíóúüïàèòçl.]{3,}\b` captures characters specific to the language, including the interpunct used in the *ll* digraph. For Spanish, the pattern is adjusted to include *ñ* and exclude Catalan-specific characters `\b[a-záéíóúüñ]{3,}\b`. In both cases, tokens shorter than three characters are discarded to prevent noise. Moreover, we designed two different language stopwords lists⁵ that are applied before keyword extraction and unigram and bigram frequency counting.

Unigram and Bigram Frequency Extraction. Unigram and bigram frequencies are computed using a single function that iterates over the stopwords-filtered token list. This produces two **Counter** objects: one mapping word forms to their frequency, and one mapping bigram strings to their co-occurrence count. The top 50 entries from each are presented in `unigrams_freq.txt` and `bigrams_freq.txt` respectively. Additionally, bigrams are computed by decade and by dialectal variant; the top 20 bigrams per group are saved to `ngrams_by_decade.txt` and `ngrams_by_variant.txt`.

Log-odds. The log-odds ratio quantifies the relative importance of a term in one period relative to another. In both single-language and comparative mode, the ratio contrasts texts published before the split year against those published from the split year onwards (default: 1920). For a word w with frequency f_A in group A (size N_A) and frequency f_B in group B (size N_B):

$$\text{Log-odds}(w) = \log \frac{P_A(w)}{1 - P_A(w)} - \log \frac{P_B(w)}{1 - P_B(w)} \quad (1)$$

Positive values indicate words over-represented in the early period; negative values indicate words distinctive of the late period. By default, a word must appear with frequency ≥ 50 in at least one group to be included. Results are written to `keywords_log_odds.txt`.

Lexical Richness. For each document group (decade and variant) the following statistics are computed over stopwords-filtered tokens:

- **Token count** and **type count**: the total number of word occurrences (tokens) and the number of distinct word forms (types) remaining in the text after stopwords removal.
- **Type-Token Ratio (TTR)**: measures lexical diversity by calculating the proportion of unique words (types) relative to the total number of words (tokens) in a text. Higher values indicate greater lexical variety, while lower values suggest more repetition. However, TTR is known to be sensitive to text length, as shorter texts tend to yield higher ratios. $\text{TTR} = \frac{\text{types}}{\text{tokens}}$.
- **Moving-Average TTR (MATTR)** (Covington and McFall, 2010): to mitigate the bias of document length, TTR is calculated over non-overlapping 500-token windows and averaged, following the recommendations of the original authors. If the group contains fewer than 500 tokens, the standard TTR is used as a fallback.
- **Hapax legomena**: words appearing exactly once, reported as a raw count and as a percentage of total types.

Comparative Metrics (--compare mode). When both language directions are analyzed together, three additional metrics are computed.

1. **Vocabulary overlap**: the set of word types shared between the Catalan and Spanish token lists (such as cognates or loanwords). It is expressed as a ratio over each language’s vocabulary.
2. **Length-ratio statistics**: calculates the expansion or contraction of text during translation by comparing token counts ($r = N_{\text{ES}}/N_{\text{CA}}$).
3. **Decade-level lexical divergence**: TTR and MATTR are compared across decades for both languages, and the difference $\Delta\text{TTR} = \text{TTR}_{\text{ES}} - \text{TTR}_{\text{CA}}$ is reported.

Output Files. Table 1 summarizes the resulting files. For each run, output files are placed under `data/analysis/linguistic/run_*/`. Per-language outputs are written to the `ca/` and `es/` subdirectories; comparative outputs go to `compare/`.

⁵ Available in the repository script `05_linguistics_analysis.py`.

Table 1. Output files produced by `05_linguistics_analysis.py`.

File	Contents
<code>ca/keywords_log_odds.txt</code>	Log-odds: texts published before 1920 vs. from 1920 onwards
<code>ca/unigrams_freq.txt</code>	Top- <i>N</i> unigrams by frequency (Catalan)
<code>ca/bigrams_freq.txt</code>	Top- <i>N</i> bigrams by frequency (Catalan)
<code>ca/ngrams_by_decade.txt</code>	Top-20 bigrams per decade (Catalan)
<code>ca/ngrams_by_variant.txt</code>	Top-20 bigrams per dialectal variant (Catalan)
<code>ca/lexical_stats.txt</code>	TTR, MATTR, hapax by decade and variant (Catalan)
<code>es/[same structure]</code>	Equivalent files for Spanish translations
<code>compare/comparative_summary.txt</code>	Cross-language overview: overlap, length ratios, divergence

5 Results

Below, we present the results obtained in the analysis. Section 5.1 comprises the results yielded by `04_analyze_corpus.py`. Subsequent sections (from 5.2 to 5.6) report the results of `05_linguistics_analysis.py`.

5.1 Corpus Composition

The final corpus comprises 63 documents totaling 1,577,899 words in the source language and 1,669,666 words in the target language, with 30,032 aligned paragraph pairs. The year of publication ranges between 1905 and 1948, with a mean of 476.7 paragraphs per document (median: 395). All texts carry the literary domain LIT classification code. Fifty-four texts are original Catalan compositions and nine are translations.

Temporal and Variant Distribution. The decade-level distribution is presented in Table 2. The 1910s are the best-represented decade with 24 documents (38.1%), followed by the 1920s (17, 27.0%), while the 1900s and 1930s each comprise ten documents (15.9%) and the 1940s two (3.2%). Average document length does not increase steadily over decades: 1910s documents average 383.5 paragraphs, rising to 537.3 in the 1920s, 559.9 in the 1930s, and 541.5 in the 1940s, while the 1900s average is 501.2.

Table 2. Corpus distribution by decade.

Decade	Documents	Avg. paragraphs
1900s	10	501.2
1910s	24	383.5
1920s	17	537.3
1930s	10	559.9
1940s	2	541.5
Mean		476.7
Median		395

As displayed in Table 3, the dialectal distribution is highly skewed towards the *central* variety (56 documents), with three *baleàric* texts, two *nord-occidental*, one *valencià*, and one *septentrional*.

Table 3. Dialectal distribution by textual variant.

Variant	Documents Share (%)	
central	56	88.9
baleàric	3	4.8
nord-occidental	2	3.2
valencià	1	1.6
septentrional	1	1.6

Table 4 shows that the most represented author is Pere Corominas with eight documents, followed by Prudenci Bertrana with seven, and Santiago Rusiñol and Joaquim Ruyra with four documents each.

Table 4. Top authors by number of documents in the corpus.

Author	Documents Share (%)	
Corominas, Pere	8	12.7
Bertrana, Prudenci	7	11.1
Rusiñol, Santiago	4	6.3
Ruyra, Joaquim	4	6.3
Bofill i Mates, Jaume	3	4.8
Clascar, Frederic	3	4.8
Vallmitjana, Juli	3	4.8

5.2 Comparative Corpus Overview

The comparative analysis reveals that the final parallel corpus contains 833,803 Catalan tokens and 840,452 Spanish tokens after postprocessing and stopword removal. The overall ES/CA token-length ratio averages 1.0080, meaning that translations tend to be slightly longer than their sources. The shared vocabulary comprises 13,166 items (15.08% of the Catalan vocabulary and 22.63% of the Spanish vocabulary). This higher overlap in Spanish may be attributed to its smaller vocabulary diversity and normalization of forms.

Table 5. Summary of comparative metrics for the aligned CA–ES corpus.

Metric	Value
Documents analysed	63
Total tokens (Catalan)	833,803
Total tokens (Spanish)	840,452
Token length ratio (ES/CA)	1.0080
Shared vocabulary (types)	13,166
Vocabulary overlap / CA	15.08%
Vocabulary overlap / ES	22.63%

5.3 N-gram Frequencies

Most Frequent Unigrams. Table 6 presents the top 20 unigrams by frequency for each language. We can note the prominence of auxiliary verbs (*havia*) and modal or light verbs (*fer*, *dir*), together with deictic demonstratives (*aquell*, *aquesta*, *aquella*). The higher frequency of *había* relative to its Catalan counterpart *havia* (5,070 vs. 3,691) may reflect the MT system’s tendency to translate Catalan periphrastic constructions into Spanish imperfect or pluperfect forms. Likewise, the strong presence of *pues* (2,541) and *bien* (2,426) among the most frequent Spanish tokens—whose Catalan near-equivalents *ara* and *ben* rank considerably lower—suggests a higher use of discourse markers and connective words in the target.

Table 6. Top 20 unigrams by frequency, Catalan and Spanish.

Rank	Catalan	Freq.	Freq.	Spanish
1	havia	3,691	5,070	había
2	cap	3,490	3,083	tan
3	fer	3,314	2,673	hombre
4	tan	3,083	2,593	ahora
5	dir	2,590	2,541	pues
6	home	2,542	2,426	bien
7	ara	2,445	2,233	casa
8	aquell	2,292	2,146	vida
9	pas	2,249	2,130	solo
10	casa	2,073	2,101	señor
11	aquesta	2,010	1,998	allí
12	aquella	1,995	1,984	día
13	vida	1,916	1,919	dos
14	tota	1,852	1,887	vez
15	ben	1,809	1,861	mientras
16	senyor	1,777	1,839	siempre
17	són	1,756	1,812	cómo
18	gran	1,714	1,809	esta
19	mateix	1,708	1,735	misma
20	terra	1,693	1,680	dios

Most Frequent Bigrams. Table 7 presents the top 20 overall bigrams for each language. The character-name bigrams shared across both languages (*maría gloria*, *rey rodolf/o*, *jordi dandin*, *conde/comte orlof*, *príncep/príncipe demetri/o*, *tomàs/tomás bajalta*) suggest that the MT system treats proper nouns as invariant and only applies slight morphophonological adaptation (*rodolf* → *rodolfo*, *tomàs* → *tomás*) when required by Spanish orthographic conventions. Grammatical collocations such as *tan sols* (only/just) do not have a direct Spanish correspondence in frequency. Notably, while the Catalan list contains only two periphrastic bigrams (*havia fet*, *havia estat*), the Spanish list includes four constructions built on *había* (*había sido*, *nunca había*, *había visto*, *había hecho*), at comparatively higher ranks. This asymmetry is consistent with the pattern noted for unigrams. During initial experiments, prompt-related bigrams were observed in the Spanish output. These appear to have been removed, as the Spanish list no longer contains (*registro tono*, *tono estilo*, *texto literario*), which confirms that the postprocessing step in `03_build_corpus.py` is effective for data cleaning.

Table 7. Top 20 bigrams by frequency, Catalan and Spanish.

Rank	Catalan	Freq.	Freq.	Spanish
1	maría gloria	215	235	mismo tiempo
2	rey rodolf	206	220	quién sabe
3	alguna cosa	198	215	maría gloria
4	don pere	190	208	todas partes
5	havia fet	186	204	rey rodolfo
6	jordi dandin	186	192	día siguiente
7	tan sols	171	187	don pedro
8	comte orlof	165	183	había sido
9	príncep demetri	158	183	jordi dandin
10	vol dir	138	167	nunca había
11	tomàs bajalta	138	164	voz baja
12	tota mena	137	163	conde orlof
13	hereu escampa	137	152	aquel hombre
14	senyor josep	135	137	había visto
15	havia estat	131	133	tal vez
16	fills israel	129	132	había hecho
17	don modest	128	130	darse cuenta
18	donya maria	121	127	doña maría
19	anna maria	116	122	tomás bajalta
20	sant pere	113	116	señor josep

Bigrams by Decade. Table 8 presents the top 5 bigrams per decade for each language, illustrating the specific thematic of the sub-corpora. In the 1900s, religious and hagionymic collocations are particularly frequent (*Sant Pere*, *Sant Roch*, *Mare de Déu*). The 1910s are mainly characterized by charactername pairs. During the 1920s, proper names such as *Tomàs Bajalta*, as well as collocations like *hereu escampa* (a squandering heir) and other fixed expressions, appear among the most frequent combinations. The 1930s and 1940s still tend to exhibit a large proportion of proper names. An interesting pattern is that in some cases the MT system does not preserve the diacritic in *Pau Garcia*.

Table 8. Top 5 bigrams per decade, Catalan and Spanish.

Decade	Rank	Catalan	Spanish
1900s	1	sant pere (81)	san pedro (76)
	2	sant roch (63)	san roque (53)
	3	mare deu (47)	todas partes (42)
	4	joan valent (43)	quién sabe (37)
	5	nostre senyor (41)	día siguiente (34)
1910s	1	maría gloria (215)	maría gloria (215)
	2	rey rodolf (206)	rey rodolfo (204)
	3	don pere (190)	don pedro (185)
	4	comte orlof (165)	conde orlof (163)
	5	príncep demetri (158)	don modesto (116)
1920s	1	tomàs bajalta (138)	tomás bajalta (122)
	2	hereu escampa (137)	hijos israel (105)
	3	fills israel (122)	heredero escampa (97)
	4	mon senyor (87)	todas partes (69)
	5	alguna cosa (83)	quién sabe (64)
1930s	1	jordi dandin (186)	jordi dandin (183)
	2	senyor sotenville (90)	señor sotenville (90)
	3	senyor marcel·lí (60)	señora mónica (53)
	4	hereu aspriu (56)	heredero aspriu (50)
	5	senyora mònica (47)	nunca había (43)
1940s	1	pau garcia (112)	anna maria (98)
	2	anna maria (112)	pau garcía (69)
	3	fra job (40)	pau garcia (44)
	4	havia estat (39)	había sido (39)
	5	bell nou (27)	fra job (37)

Bigrams by Dialectal Variant. Table 9 displays the top five bigrams for each dialectal variant. The *central* variety mirrors the general corpus distribution due to its 88.9% dominance, whereas the peripheral variants have some more specific lexicon. The *septentrional* and *valencià* subsets are characterized by devotional terms (*Sant Pere* (Saint Peter), *Mare de Déu* (Virgin Mary)). Conversely, the *nord-occidental* is shaped by the presence of Shakespeare works, as evidenced by names like *Hàmlet*. Finally, the *baleàric* sub-corpus is distinguished by its essayistic discourse, with markers such as *tal volta* (perhaps) or *mejor dicho* (more precisely).

Table 9. Top 5 bigrams per dialectal variant, Catalan and Spanish.

Variant	Rank	Catalan	Spanish
baleàric	1	tal volta (10)	tal vez (5)
	2	pera fer (8)	mejor dicho (5)
	3	bon mot (5)	nuestra lengua (5)
	4	hora present (5)	amor propio (4)
	5	sos ulls (4)	pecar lengua (4)
central	1	maría gloria (215)	mismo tiempo (217)
	2	rey rodolf (206)	maría gloria (215)
	3	don pere (190)	rey rodolfo (204)
	4	jordi dandin (186)	todas partes (187)
	5	alguna cosa (182)	don pedro (185)
nord-occ.	1	mon senyor (87)	doña ramona (41)
	2	donya ramona (40)	señor hamlet (38)
	3	senyor hâmllet (38)	doña maría (31)
	4	donya maria (32)	nunca había (30)
	5	primer fosser (30)	rosencrantz guildenstern (23)
septentrional	1	sant pere (77)	san pedro (74)
	2	sant roch (63)	san roque (53)
	3	joan valent (43)	nuestro señor (26)
	4	mare deu (40)	señor rector (26)
	5	nostre senyor (38)	joan valent (24)
valencià	1	sant roc (28)	san roque (22)
	2	mare déu (22)	quién sabe (16)
	3	los ulls (15)	niño jesús (16)
	4	mestre domingo (15)	maestro domingo (15)
	5	seua filla (14)	señor batiste (15)

5.4 Log-odds Ratio

The log-odds ratio identifies terms significantly over-represented in one period relative to the other. We contrasted texts published before 1920 against those published from 1920 onwards—a threshold chosen to capture the orthographic transition following the publication of the *Normes ortogràfiques* (Institut d’Estudis Catalans, 1913), which standardized written Catalan from 1913 onwards. The top 10 terms for each period and language are shown in Table 10.

Table 10. Top 10 log-odds terms per period and language (early <1920; late $\geq$ 1920).

Catalan (CA)			Spanish (ES)		
Rank	Term	Score	Rank	Term	Score
<i>Early period (<1920) — positive log-odds</i>					
1	<i>havía</i>	7.84	1	<i>gladys</i>	6.87
2	<i>perque</i>	7.07	2	<i>enriqueta</i>	6.69
3	<i>séu</i>	6.95	3	<i>nataniel</i>	6.60
4	<i>gladys</i>	6.87	4	<i>rodolfo</i>	6.35
5	<i>may</i>	6.79	5	<i>orlof</i>	6.23
6	<i>tenía</i>	6.73	6	<i>andrónica</i>	6.21
7	<i>rès</i>	6.72	7	<i>tarregada</i>	6.10
8	<i>enriqueta</i>	6.67	8	<i>nicéforo</i>	5.95
9	<i>maría</i>	6.62	9	<i>demetrio</i>	5.93
10	<i>vareig</i>	6.60	10	<i>heráclias</i>	5.77
<i>Late period ($\geq$1920) — negative log-odds</i>					
1	<i>innocenci</i>	-7.22	1	<i>aspriu</i>	-6.92
2	<i>aspriu</i>	-7.01	2	<i>innocenci</i>	-6.69
3	<i>hàmlet</i>	-6.41	3	<i>ernestina</i>	-6.32
4	<i>ernestina</i>	-6.35	4	<i>inocencio</i>	-6.10
5	<i>eugènia</i>	-6.17	5	<i>bajalta</i>	-6.08
6	<i>bajalta</i>	-6.10	6	<i>merceneta</i>	-5.68
7	<i>júlia</i>	-5.99	7	<i>dandín</i>	-5.67
8	<i>lluïsa</i>	-5.90	8	<i>temme</i>	-5.49
9	<i>sirus</i>	-5.86	9	<i>sirus</i>	-5.47
10	<i>cremes</i>	-5.77	10	<i>júlia</i>	-5.45

Catalan log-odds. The early period is dominated by two distinct types of terms. The first consists of character names tied to specific works of that period (*gladys*, *enriqueta*, *nataniel*, *rodolf*, *orlof*), which do not reflect any linguistic property. The second, more linguistically meaningful, group consists of pre-normative orthographic variants that were standardized by the reforms of 1913: *havía*, *tenía*, *feya*, *deya* (later *havia*, *tenia*, *feia*, *deia*); *donchs*, *axò*, *avuy*, *lloch*, *hont*, *ahont*, *aygua*, *poch*, *xich* (later *doncs*, *això*, *avui*, *lloc*, *on*, *on*, *aigua*, *poc*, *xic*); and synthetic verb forms such as *tinch*, *puch*, *vareig* (later *tinc*, *puc*, *vaig*). The presence of these forms confirms that the corpus captures the orthographic transition undergone by written Catalan in the early twentieth century. The late period is similarly dominated by character names from works concentrated in that period, with only a handful of genuinely modern lexical items (*tanmateix*, *prendre*, *vore*) appearing among the top-ranked terms.

Spanish log-odds. The Spanish translations show a strikingly different picture. Both the early and the late period are dominated almost entirely by character names, with no orthographic or morphological variation of the kind observed in the Catalan originals. This suggests that the translation process neutralizes the diachronic variation present in the source texts and its style is more homogeneous.

5.5 Lexical Richness

Tables 11, 12 and 13 present lexical richness statistics overall, by decade, and by variant for Catalan and Spanish, respectively.

Hapax Legomena. As shown in Table 11, Catalan exhibits a higher overall hapax rate (46.9%) than Spanish (38.4%).

Tables 12 and 13 further show that hapax values are consistently higher in Catalan than in Spanish across all decade groups (e.g., 1900s: 58.0% vs. 50.1%; 1910s: 52.9% vs. 44.6%).

The *baleàric* variant exhibits the highest hapax percentages in both languages (CA: 68.4%; ES: 66.8%), which is consistent with its small token count and distinctive lexicon. The *central* variety has the lowest hapax ratios (CA: 46.6%; ES: 38.9%), attributable to its much larger size and the law-of-large-numbers effect on hapax.

Table 11. Overall lexical richness statistics for Catalan and Spanish.

Language	Tokens	Types	TTR	MATTR	Hapax	Hapax%
Catalan (CA)	833,803	87,335	0.1047	0.7673	40,954	46.9%
Spanish (ES)	840,452	58,173	0.0692	0.7907	22,334	38.4%

Table 12. Lexical richness statistics for Catalan (by decade and variant).

Group	Tokens	Types	TTR	MATTR	Hapax	Hapax%
<i>By decade</i>						
1900s	117,476	27,034	0.2301	0.7570	15,692	58.0%
1910s	252,804	43,847	0.1734	0.7498	23,202	52.9%
1920s	243,317	37,300	0.1533	0.7506	18,383	49.3%
1930s	152,777	32,255	0.2111	0.7868	17,071	52.9%
1940s	67,429	19,179	0.2844	0.8597	10,657	55.6%
<i>By variant</i>						
baleàric	14,356	6,726	0.4685	0.8301	4,600	68.4%
central	724,131	79,486	0.1098	0.7685	37,074	46.6%
nord-occidental	39,160	9,560	0.2441	0.7469	5,440	56.9%
septentrional	25,514	5,774	0.2263	0.6715	3,238	56.1%
valencià	30,642	9,971	0.3254	0.7893	5,870	58.9%

Table 13. Lexical richness statistics for Spanish translations (by decade and variant).

Group	Tokens	Types	TTR	MATTR	Hapax	Hapax%
<i>By decade</i>						
1900s	114,819	21,534	0.1875	0.7926	10,787	50.1%
1910s	253,048	32,968	0.1303	0.7712	14,705	44.6%
1920s	250,790	32,519	0.1297	0.7823	14,563	44.8%
1930s	154,455	27,026	0.1750	0.8052	13,056	48.3%
1940s	67,340	16,977	0.2521	0.8594	8,876	52.3%
<i>By variant</i>						
baleàric	13,527	6,253	0.4623	0.8292	4,175	66.8%
central	729,860	54,953	0.0753	0.7911	21,357	38.9%
nord-occidental	40,450	10,313	0.2550	0.7763	5,759	55.8%
septentrional	24,343	6,753	0.2774	0.7494	3,871	57.3%
valencià	32,272	9,880	0.3061	0.8144	5,587	56.5%

TTR and MATTR. As expected, TTR values are inversely correlated with corpus size across all groups, reflecting the well-known length dependence of this metric. MATTR mitigates this bias and provides a more stable estimate. Contrary to the initial hypothesis—stated in Section 1—that LLM output would normalize vocabulary and thus produce lower lexical diversity than the source texts, Spanish MATTR scores are higher **overall** (0.7907 vs. 0.7673) and also in **most decades**, with the 1940s as the only exception. The gap is most pronounced in the 1900s (0.0356) and decreases to near parity by the 1940s. This outcome may reflect the fact that Gemma3-27B-IT, as an LLM, has a broad productive vocabulary range. As regards the **variant-level MATTR** scores, they show the *baleàric* sub-corpus as the highest in both languages (CA: 0.8301; ES: 0.8292), expected due to its small size, highly distinctive morphology, and essayistic register. The *septentrional* variety scores lowest in Catalan (0.6715) and in Spanish (0.7494), which may be driven by its older orthographic conventions (since it contains only one text with a publication date of 1907). The *central* variety, by far the largest sub-corpus, scores 0.7685 (CA) and 0.7911 (ES).

Decade-level Lexical Divergence. Table 14 reports the TTR difference $\Delta\text{TTR} = \text{TTR}_{\text{ES}} - \text{TTR}_{\text{CA}}$ and the MATTR values for each decade. The ΔTTR is negative in all decades; Spanish translations consistently display a lower type-token ratio than their Catalan sources. The largest difference is located in the 1910s (-0.0431) and the smallest in the 1920s (-0.0236). Regarding MATTR, Spanish shows slightly higher values in most decades (1900s–1930s), with the sole exception of the 1940s, where Catalan marginally exceeds Spanish (0.8597 vs. 0.8594). This leads us to conclude that, although Spanish has a more limited vocabulary, its lexical distribution is more locally even across sliding windows.

Table 14. Decade-level lexical divergence between Catalan and Spanish.

Decade	CA TTR	ES TTR	ΔTTR	CA MATTR	ES MATTR
1900s	0.2301	0.1875	-0.0426	0.7570	0.7926
1910s	0.1734	0.1303	-0.0431	0.7498	0.7712
1920s	0.1533	0.1297	-0.0236	0.7506	0.7823
1930s	0.2111	0.1750	-0.0361	0.7868	0.8052
1940s	0.2844	0.2521	-0.0323	0.8597	0.8594

Overall, we observe a mismatch between different measures of lexical diversity. Catalan exhibits higher hapax rates and a greater number of types, but Spanish shows slightly higher MATTR values. Our interpretation is that hapax and TTR are more sensitive to low-frequency and morphologically rich forms, whereas MATTR captures a more stable notion of diversity and is less influenced by rare tokens. Along the same lines, in the target texts, the MT system appears to normalize rare forms towards a more uniform vocabulary, thereby reducing the hapax ratio. **In essence, Catalan is richer in morphological variety, whereas the Spanish translations show a more even local distribution.**

5.6 Document Length Ratios

The per-document ES/CA length ratio reveals that all 63 translated documents are longer than their sources across the entire corpus (Table 15). The highest ratio is observed for document *2660* (**1.1617**), followed by *2604* (1.1407), *2601* (1.1182), and *2873* (1.1161). This expansion is also reflected in the mean document-level ratio of **1.0080** across all 63 documents and aligns with the tendencies reported in Section 5.3—namely, the LLM’s tendency to introduce additional connectives and explicitations.

Table 15. Top 10 documents by ES/CA length ratio ($N_{\text{ES}}/N_{\text{CA}}$ tokens).

Doc ID	Year	tokens (CA)	tokens (ES)	Ratio
2660	1929	15,260	17,728	1.1617
2604	1920	5,897	6,727	1.1407
2601	1922	2,505	2,801	1.1182
2873	1915	4,392	4,902	1.1161
0378	1935	40,048	44,371	1.1079
3259	1908	11,001	12,126	1.1023
0154	1927	2,810	3,075	1.0943
3309	1905	10,459	11,418	1.0917
3087	1906	3,105	3,388	1.0911
0098	1922	36,437	39,556	1.0856
Overall mean				1.0080

6 Discussion

6.1 Design Decisions

The corpus is restricted to post-1905 texts to avoid pre-standard Catalan forms and to ensure a more homogeneous literary context for diachronic comparison.

The selection of Gemma3-27B-IT as the translation engine, rather than a dedicated smaller MT model such as MarianMT, is motivated by the LLM’s known competitive performance in producing more idiomatic and fluent texts, a requirement given the nature of the literary domain.

6.2 Limitations

Several limitations should be borne in mind when interpreting the results.

First, all translations are produced by a single MT system, meaning that observations on the Spanish side may reflect system-specific biases. Second, the corpus is unbalanced across decades and variants: the 1940s sub-corpus consists of only two documents, and minority varieties (*baleàric*, *nord-occidental*, *valencià*, *septentrional*) are each represented by one to three texts. Third, the tokenization uses simple regular expressions rather than morphological analyzers, which may slightly overcount types for morphologically productive forms. Fourth, the pipeline relies on fully automatic alignment and preprocessing, which may introduce minor inaccuracies in edge cases.

6.3 Reflections on the Pipeline

The pipeline proved well suited for the corpus construction and the subsequent analysis. The filtered metadata CSV file provides a registry linking all analyses to the original source documents for experimentation. In addition, the alignment procedure preserves traceability throughout the pipeline and ensures that translations remain matchable to their source segments via offsets. The separation between distribution analysis (*Script 4*) and linguistic analysis (*Script 5*) enables independent execution after translation, with flexible parameters (e.g. top- N , frequency thresholds, and filtering by year or variant). Furthermore, the removal of MT produced markers by *Script 3* enabled a clean comparative analysis and a high-quality parallel corpus.

7 Conclusions

This paper has presented the construction and analysis of a synthetic parallel corpus of early twentieth-century Catalan literary texts and their automatic Spanish translations generated by an LLM. The five-script pipeline encompasses data loading and normalization, machine translation, aligned corpus construction with artifact removal, and corpus distribution and extended linguistic analysis. The result is a 63-document resource comprising 1,577,899 Catalan words and 1,669,666 Spanish words, with 30,032 aligned paragraph pairs spanning the period 1905–1948 across five dialectal variants of Catalan.

The linguistic analysis reveals several consistent findings. Hapax percentages are higher in Catalan than in Spanish across all decade groups, reflecting both the richer morphological variation of literary Catalan and the normalizing effect of the MT system. MATTR scores are broadly comparable across the two languages, with small but consistent advantages in the Spanish direction, leading us to conclude that Gemma3-27B-IT does not drastically reduce lexical diversity relative to the source. The log-odds analysis highlights the diachronic variation attested in the Catalan texts. In contrast, the Spanish translations show no comparable orthographic or morphological differentiation across periods, suggesting that the translation process neutralizes the lexical variation present in the source texts.

The bigram analysis confirms that named entities and fixed cultural references are preserved as invariant translation units. Document-level length ratios cluster consistently above 1.0 (mean 1.0080), reflecting a general tendency for translations to be slightly longer than the source texts.

Future work could extend the corpus to include human translations where available for direct comparison with the MT output, broaden the temporal scope to post-1950 texts, and apply lemmatization and part-of-speech tagging for syntactic comparison alongside the lexical analyses presented here. A further consideration would be the integration of a more balanced distribution of texts across Catalan varieties—particularly *baleàric*, *nord-occidental*, *septentrional*, and *valencià*—which would further strengthen the analysis and support more reliable comparisons in the less-resourced varieties.

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A Literary Corpus

This appendix lists the 63 literary works included in the corpus generated in this study. All texts are drawn from the *Corpus Textual Informatitzat de la Llengua Catalana* (CTILC). Works are ordered alphabetically by author's surname. The column *Variant* refers to the dialectal variety of Catalan: *Central*, *Balearic*, *North-Western*, *Northern*, and *Valencian*. The column *Transl.* indicates whether the text is a translation (Y) or an original work (N).

Author	Title	Year	Variant	Transl.
Albert i Pey, Salvador	<i>Involució</i>	1908	Central	N
Alcover, Joan	<i>Cultura de llenguatge</i>	1916	Balearic	N
Alcover, Joan	<i>Poemes bíblics</i>	1918	Balearic	N
Alomar, Gabriel	<i>El futurisme</i>	1905	Balearic	N
Bertrana, Prudenci	<i>Josafat</i>	1906	Central	N
Bertrana, Prudenci	<i>Les ales d'Ernestina</i>	1921	Central	N
Bertrana, Prudenci	<i>El meu amic Pellini</i>	1923	Central	N
Bertrana, Prudenci	<i>El fantasma de Montcorb</i>	1930	Central	N
Bertrana, Prudenci	<i>L'hereu</i>	1931	Central	N
Bertrana, Prudenci	<i>L'ós benemèrit i altres bèsties</i>	1932	Central	N
Bertrana, Prudenci	<i>L'impenitent</i>	1948	Central	N
Bofill i Mates, Jaume	<i>Selvatana amor</i>	1920	Central	N
Bofill i Mates, Jaume	<i>Sàtires</i>	1929	Central	N
Bofill i Mates, Jaume	<i>L'altra concòrdia</i>	1930	Central	N
Capdevila, Carles	<i>L'home i les armes</i>	1934	Central	Y
Careta i Vidal, Antoni	<i>Narracions estranyes</i>	1905	Central	N
Casellas, Raimon	<i>Etapes estètiques</i>	1916	Central	N
Casponce, Esteve	<i>Contes vallespirenchs</i>	1907	Northern	N
Clascar, Frederic	<i>El Gènesi</i>	1914	Central	Y
Clascar, Frederic	<i>El Càntic dels Càntics de Salomó</i>	1918	Central	Y
Clascar, Frederic	<i>L'Èxode</i>	1925	Central	Y
Collell, Jaume	<i>Memories d'un noy de Vich</i>	1908	Central	N
Corominas, Pere	<i>Les albes de la vida</i>	1908	Central	N
Corominas, Pere	<i>Les gràcies de l'Empordà</i>	1919	Central	N
Corominas, Pere	<i>Jardins de Sant Pol</i>	1927	Central	N
Corominas, Pere	<i>De plaer no n'hi ha mai prou</i>	1927	Central	N
Corominas, Pere	<i>Les dites i facècies de l'estrenu filàntrop en Tomàs de Bajalta. II, Pigmalió</i>	1928	Central	N
Corominas, Pere	<i>Pina, la italiana del dancing</i>	1933	Central	N
Corominas, Pere	<i>El botrí de si mateix</i>	1936	Central	Y
Corominas, Pere	<i>La Rambla</i>	1940	Central	N
Crexells, Joan	<i>Diàlegs, I</i>	1924	Central	Y
Folguera, Joaquim	<i>El poema espars</i>	1917	Central	N
Gener, Pompeu	<i>La taverna intel·lectual</i>	1917	Central	N
Guimerà, Àngel	<i>Andrònica</i>	1905	Central	N
Guimerà, Àngel	<i>Jesús que torna</i>	1917	Central	N
Iglésies i Guizard, Josep	<i>Segarrenques</i>	1917	Central	N
Karr i Alfonsetti, Carme	<i>Primera conferència</i>	1910	Central	N
Karr i Alfonsetti, Carme	<i>Els ídols</i>	1911	Central	N
Maragall, Joan	<i>Elogis</i>	1912	Central	N
Maseras i Galtés, Alfons	<i>La intel·ligència de les flors</i>	1930	Central	Y
Maseras i Galtés, Alfons	<i>Jordi Dandin, o el marit confús</i>	1932	Central	Y
Massó i Torrents, Jaume	<i>Croquis pirenenacs</i>	1921	Central	N

Author	Title	Year	Variant	Transl.
Monserdà, Dolors	<i>María Gloria</i>	1917	Central	N
Morató i Grau, Josep	<i>El tresor</i>	1912	Central	N
Morera Galícia, Magí	<i>Hàmlet</i>	1920	North-Western	Y
Oller, Narcís	<i>Rurals y urbanes</i>	1916	Central	N
Oller, Narcís	<i>Al llapis y a la ploma</i>	1918	Central	N
Pascual Tirado, Josep	<i>De la meua garbera</i>	1935	Valencian	N
Plana, Alexandre	<i>A l'ombra de Santa Maria del Mar</i>	1923	North-Western	N
Poal i Aragall, Miquel	<i>Ço que deu ésser el feminisme</i>	1915	Central	N
Rusiñol, Santiago	<i>El senyor Josep falta a la dòna</i>	1915	Central	N
Rusiñol, Santiago	<i>En Josepet de Sant Celoni</i>	1917	Central	N
Rusiñol, Santiago	<i>L'illa de la calma</i>	1922	Central	N
Rusiñol, Santiago	<i>Màximes i mals pensaments</i>	1927	Central	N
Ruyra, Joaquim	<i>El país del Pler</i>	1906	Central	N
Ruyra, Joaquim	<i>La parada</i>	1919	Central	N
Ruyra, Joaquim	<i>L'idil·li d'en Temme</i>	1920	Central	N
Ruyra, Joaquim	<i>Les coses benignes</i>	1925	Central	N
Salvat-Papasseit, Joan	<i>La gesta dels estels</i>	1922	Central	N
Vallmitjana, Juli	<i>Sota Montjuic</i>	1908	Central	N
Vallmitjana, Juli	<i>L'abella perduda</i>	1910	Central	N
Vallmitjana, Juli	<i>Els zin-calós</i>	1911	Central	N
Vidal i Rosich, Plàcid	<i>Somni de primavera</i>	1911	Central	N